Module 4: Critical Thinking

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Critical Thinking Assignment (75 Points)

Simulate the First-Fit Algorithm in Allocating Memory to Processes

In memory management, the First-Fit Algorithm allocates the first available memory block that can accommodate a process. Your task is to create a program that simulates the First-Fit Algorithm for memory allocation in an operating system using a programming language of your choice. Test your algorithm with varying memory block sizes and process sizes.

In your summary paper:

- Explain how the First-Fit Algorithm works and compare it with the Best-Fit Algorithm.
  - Include screenshots of the source code in your paper for reference.
- Can you think of situations where the Best-Fit Algorithm is preferable to the First-Fit Algorithm? Explain your reasons for that opinion.

Grading Criteria

- Your paper should be 2-3 pages in length, not including the cover page and references page.
- Your paper must be formatted according to APA guidelines in the CSU Global Writing Center (available in the left-hand navigation panel).
- Your claims should be supported by evidence. Include at least 3 credible references. The CSU Global Library (available in the left-hand navigation panel) is a good place to find these references.
- All references must be cited in the text and listed on the references page, according to APA formatting.
- See the rubric below for more details about how you will be graded for this assignment.

CSC507_M04_CT

Criteria	Ratings				Pts
Program Implementation and Functionality	**20 pts** **Meets Expectation** Program correctly implements the First-Fit Algorithm. Demonstrates accurate simulation of memory allocation with varying memory block and process sizes. Outputs are correct and clearly documented.	**16 pts** **Approaches Expectation** Program mostly implements the First-Fit Algorithm but may contain minor logic or testing errors. Results are generally correct but lack full verification or clarity.	**12 pts** **Below Expectation** Program attempts implementation but contains significant logic errors or fails to test with varying sizes. Results are incomplete or inaccurate.	**8 pts** **Limited Evidence** Program missing or does not compile/run. No clear evidence of understanding First-Fit implementation.	20 pts
Explanation of the First-Fit Algorithm	**15 pts** **Meets Expectation** Provides a clear, accurate, and detailed explanation of how the First-Fit Algorithm works, including how it allocates memory and handles fragmentation. Demonstrates strong conceptual understanding.	**12 pts** **Approaches Expectation** Explanation is mostly correct but lacks depth or omits minor details about algorithm operation or fragmentation.	**8 pts** **Below Expectation** Explanation shows limited understanding; key concepts or terms are missing or misrepresented.	**4 pts** **Limited Evidence** Explanation absent or incorrect; shows little understanding of the algorithm.	15 pts
Comparison with Best-Fit Algorithm	**10 pts** **Meets Expectation** Provides a clear, insightful comparison between First-Fit and Best-Fit algorithms, addressing	**8 pts** **Approaches Expectation** Comparison is generally accurate but lacks full analysis or depth in one or more areas.	**6 pts** **Below Expectation** Comparison is minimal or contains inaccuracies about one or both algorithms.	**4 pts** **Limited Evidence** No comparison or only vague statements with no technical content.	10 pts

Criteria	Ratings				Pts
	efficiency, fragmentation, and performance trade-offs.				
Critical Analysis: Situations Favoring Best-Fit	**10 pts** **Meets Expectation** Thoughtfully identifies and explains situations where Best-Fit would be preferable, providing logical reasoning supported by examples or evidence.	**8 pts** **Approaches Expectation** Provides some reasoning or examples, but explanations lack clarity or depth.	**6 pts** **Below Expectation** Mentions scenarios but with little justification or evidence.	**4 pts** **Limited Evidence** No discussion or unsupported assertions.	10 pts
Inclusion and Clarity of Source Code Screenshots	**5 pts** **Meets Expectation** Screenshots of source code are clear, well-labeled, and effectively integrated into the paper. Code readability and structure demonstrate professional presentation.	**4 pts** **Approaches Expectation** Screenshots included but lack labeling, clarity, or connection to discussion.	**3 pts** **Below Expectation** Screenshots are incomplete, hard to read, or missing critical sections.	**2 pts** **Limited Evidence** No source code screenshots included.	5 pts
Use of Sources and Evidence	**10 pts** **Meets Expectation** Includes at least 3 credible references. Sources are relevant, well-integrated, and	**8 pts** **Approaches Expectation** Includes 2–3 references, but some may lack credibility,	**6 pts** **Below Expectation** Includes fewer than 2 credible references or incomplete citation formatting.	**4 pts** **Limited Evidence** No credible references or evidence provided; citations missing.	10 pts

Criteria	Ratings				Pts
	support arguments effectively. All references cited in proper APA style.	relevance, or proper citation.			
Organization, Clarity, and APA Formatting	**5 pts** **Meets Expectation** Paper is 2–3 pages (excluding cover and references). Well-organized, logically structured, and clearly written with minimal grammar or spelling errors. Follows APA formatting throughout.	**4 pts** **Approaches Expectation** Paper meets most length and structure requirements. Minor APA or grammar errors do not impede understanding.	**3 pts** **Below Expectation** Paper lacks clear structure or has significant formatting/grammar issues. APA errors present.	**2 pts** **Limited Evidence** Paper poorly organized, significantly below length, or lacks APA formatting.	5 pts
					Total Points: 75